뉴턴 및 점탄성 유체 공동흐름을 이용한 박테리아 제거

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Bacteria isolation using co-flow of Newtonian and viscoelastic fluids

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Abstract

Elasto-inertial microfluidics with a co-flow of Newtonian and viscoelastic fluids has garnered significant attention due to improved separation efficiency, resolution and throughput compared to the conventional inertial microfluidic approaches only with Newtonian or viscoelastic fluids. Recently, we suggest that w_n/d, where w_n is the Newtonian fluid width, d is the particle diameter, can serve the role of the blockage ratio of elsto-inertial microfluidics. In this paper, we propose a theoretical model to explain the behavior of particles in the co-flow of Newtonian and viscoelastic fluids based on the w_n/d value obtained through co-flow controlled particle focusing technique. This model is further validated by examing the critical w_n/d value across channels with various dimensions. Subsequently, we achieve the particle separation with submicron resolution using the proposed theoretical model. Finally, we demonstrate the direct isolation of bacteria from the sample fluid, highlighting the potential of elasto-inertial microfluidics for high-precision biological particle sorting.

Keywords: **Elasto-inertial microfluidics (탄성관성 미세유체역학), Bacteria removal (박테리아 제거), Viscoelastic fluid (점탄성 유체)**

1. Introduction

Microfluidic particle manipulation has emerged as a promising alternative to the conventional methods for sample preparation, especially for biomedical and biological assays, for sorting, separation, focusing, trapping, and washing of the bioparticles including cells, bacteria, and microalgae. Inertial microfluidics relies solely on hydrodynamic forces and thus offers various advantages including simplicity in device fabrication and operation, high-throughput, and scalability, compared to active microfluidic approaches with acoustic, electric, magnetic, and optical force fields. The inertial lift force (F_{IL}) arising inside the microchannel and depends on the particle size (d), which is wall-induced lift force ($F_W \propto d^6$) and shear-gradient lift force ($F_S \propto d^3$).[1] Several following studies have significantly deepened the understanding of physics underlying this interesting hydrodynamic phenomenon. As the inertial lateral migration velocity is proportional to the particle size, microparticles with different sizes can be separated using the inertial microfluidic approach. For further advancements in efficiency and throughput, viscoelastic fluids have been introduced to induce an additional elastic lift force ($F_{EL} \propto d^3$) in elasto-inertial microfluidics.

The microchannel cross-section dimension is a crucial factor in (elasto-)inertial microfluidic particle manipulation as the F_{IL} and F_{EL} are dependent on the channel dimension. Blockage ratio defined as $\kappa = d/D_h$, where D_h is hydraulic diameter of the microchannel.[2] The recent studies reported that κ should be greater than 0.06-0.07 for successful inertial particle focusing and manipulation in finite length channels.[3] With increasing κ, both F_{IL} and F_{EL} experienced by the particles within the channel also increase. In other words, smaller channels should be used for small particle manipoulation while the larger particles require larger microchannels for inertial microfluidic manipulation.The blockage ratio has been widely used as a design rule for single fluid-based (elasto-)inertial microfluidics with both Newtonian and viscoelastic fluids. However, the blockage ratio cannot be used directly for elasto-inertial microfluidics based on a co-flow of Newtonian and viscoelastic fluids due to the presence of the fluid interface.

A Finding on theoretical value of dimensionless number ratio between Newtonian width over particle size (w_n/d), opens a broad picture of how the prediction of focusing regimes can occur for spherical particle.[4] From a perspective of the blockage ratio, our research can be directed to elasto-inertial microfluidics to manipulate micro/nano scale objects (e.g., bacteria) in micro/nanochannels whose characteristic dimensions are smaller than those available so far (usually O(~10–100 μm)).

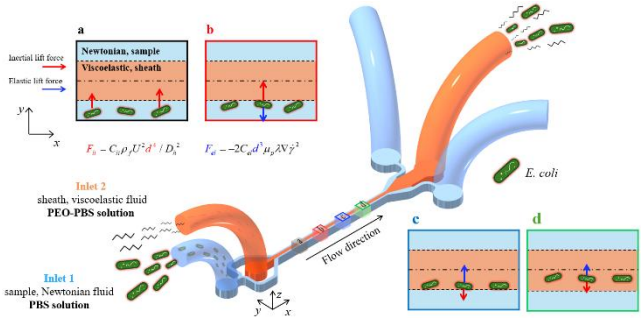

Fig. 1 Schematic of a microfluidic device for bacteria isolation using co-flow of Newtonian and viscoelastic fluids. A Newtonian sample fluid (sky blue, phosphate buffer solution) containing suspended bacteria and the viscoelastic sheath fluid (orange, PEO solution) were introduced through inlets 1 and 2, respectively.

2. Results and Discussion

2.1 Dimensionless analysis for the prediction of particle trajectories

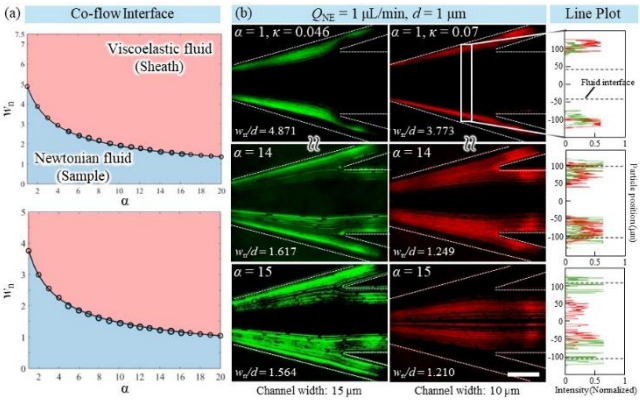

Fig. 2 (a) Newtonian fluid width w_n as a function of α = 1–20. The line represents the theoretical prediction of w_n in a fully developed two-dimensional Poiseuille flow. (b) Fluorescence trajectories of 1 μm microspheres in channels with width of 15 μm (green) and 10 μm (red) under various a conditions. The graphs indicate the particle positions measured 520 μm away from the expansion channel. Scale bar = 100 μm.

Fig. 2 (a) shows the Newtonian width w_n, while the graphs above and below show the interface position in a rectangular channel with identical aspect ratios and widths of 15 μm and 10 μm, respectively. The interface position determined by the sheath flow rate (Q_{VE}) over the sample flow rate (Q_{NE}), denoted as α. The graph demonstrates that as α increases, the variation of w_n with respect to the flow rate ratio gradually decreases. This indicates that the proposed w_n/d exhibits a critical threshold depending on the channel width, suggesting that the channel dimensions should be optimized based on the size of the target particles for effect separation.

To investigate the particle migration behavior near the co-flow interface through dimensionless analysis, 1 μm spherical has been performed under varying α. This particle size was chosen to mimic the bacteria (e.g., *E. coli*, *S. aureus*). At α = 15, unlike the particle trajectories in the 15 μm-wide channel, all 1 μm particles in the 10 μm-wide channel successfully cross the interface and separate. The w_n/d value at the point where all particles can laterally migrate across the interface is calculated to be 1.21, which is a challenging threshold to reach in the 15 μm-wide channel. [Fig. 2 (b)]

2.2. Direct isolation of *E. coli*

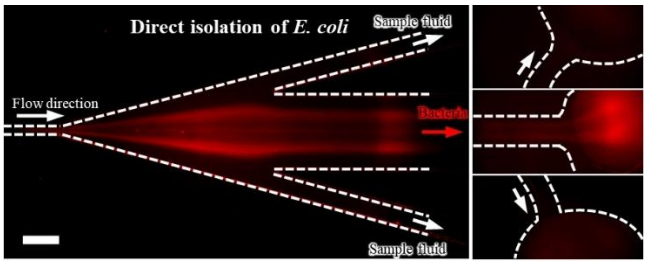

Fig. 3 The trajectories of *E. coli* under volumetric flow rate condition of Q_n = 1 μL min^{-1} and Q_{ve} = 15 μL min^{-1}. *E. coli* move toward the center outlet. Scale bar = 100 μm

Based on the successful isolation of the 1 μm particles, we further apply the present technique to the isolation of *E. coli* (1–2 μm). As both F_W and F_{EL} act toward the microchannel center and are proportional to the particle size, it can be confirmed that almost all *E. coli* are isolated from the sample fluid.

The number density of *E. coli* can be represented by the length fraction (L_f). The L_f serves as a useful descriptor when *E. coli* are tightly focused into a single stream. In this case, the value of L_f provides a clear measure of how separated the particles are on average within the channel. Under these experimental conditions, the throughput for the case with a length fraction (L_f = 0.19) is calculated to be 1667 particles/s, which is approximately more than four times higher than that of a conventional inertial microfluidic method. It implies that this isolation method enables dilution-free processing.

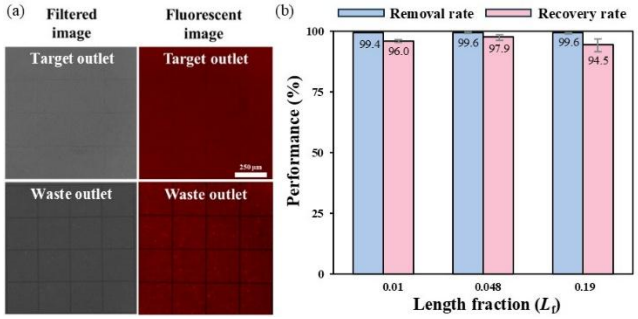

Fig. 4 (a) The hemocytometer images of *E. coli* collected from the target(side) outlet and waste(center) outlet. (b) Bacteria removal efficiency: removal rates and recovery rates over Lf values from 0.01 to 0.19. Scale bar = 50 μm

3. Conclusion

We introduced an analysis based on dimensionless number w_n/d to determine the optimal channel dimensions for separating 1 μm particles and achieve direct isolation of bacteria. This result highlights the importance of channel dimensions, particularly the influence of channel width on the critical point of w_n in Newtonian/viscoelastic co-flow. Additionally, through the experimental validation of E. coli separation, we confirmed that the direct isolation method enables dilution-free and high-throughput processing. By this evidence, we assume that an appropriate channel dimension can be established according to the target particle size, along with a theoretical model to predict particle behavior in the co-flow of Newtonian and viscoelastic fluids. To verify our hypothesis, we are gathering additional evidence using the dimensionless number w_n/d.

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